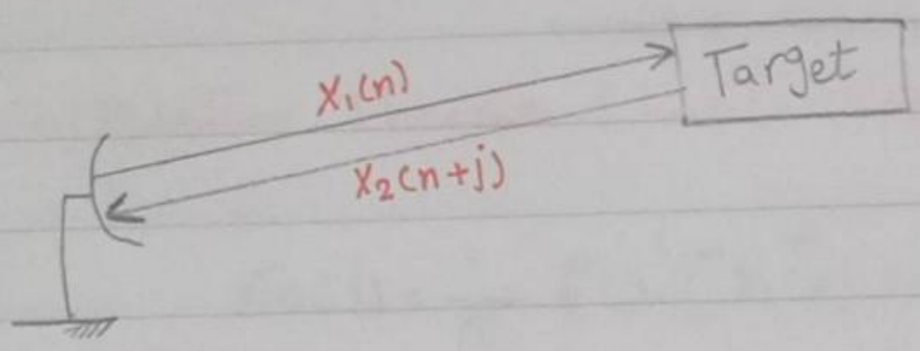
Chapter #3 Correlation & Convolution

→ Correlation :-

Consider How two data sequences, each consisting of sampled values Taken From the Two corresponding waveform, might be Compared To extracting information

$$r_{12}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)\, x_2(n+j), \quad j = 0 \dots N-1 = \text{Lag} = \text{Time delay}$$



Types of Correlation

$x_1(n) = x_2(n)$

Auto Correlation

$$r_{11}(j) = \frac{1}{N} \sum x_1(n)\, x_1(n+j)$$

⇒ Shift Left only

Auto Correlation: is The Special Case of cross-correlation, it's occure when $x_1(n) = x_2(n)$

$x_1(n) \neq x_2(n)$

cross-Correlation

$N_1 = N_2$

circular C.C

shift left + Rotate

$N_1 \neq N_2$

Linear

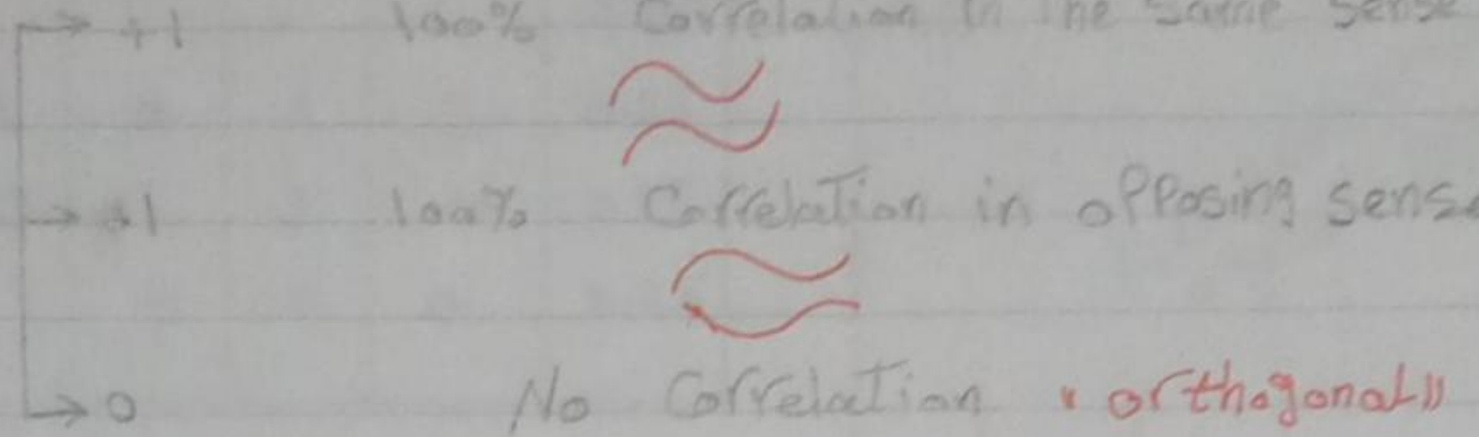
$N_{new} = N_1 + N_2 - 1$

shift left + Rotate

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Cross - Correlation Coefficient: $\rho_{12}(j)$

Its Value always Lies between -1 and +1



$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\sqrt{\sum_{n=0}^{N-1} x_1(n)^2 \cdot \sum_{n=0}^{N-1} x_2(n)^2}} \qquad -1 \leq \rho_{12}(j) \leq 1$$

---

* Correlation Theorem (Fast - Correlation)

$$r_{12}(j) = \frac{1}{N} FD^{-1}\left[X_1^*(k)\, X_2(k)\right]$$

---

① use The Direct method For data sequences

$$X_1(n) = \{1, 1, 0, 1\}$$
$$X_2(n) = \{1, 0, 0, 1\}$$

To Find

1-* Auto Correlation For each signal
* cross - Correlation
* cross - Correlation Coefficient
* Plot cross - Correlation against Lag

Solution

1-* Auto Correlation Foreach signal

$X_1(n) = \{1, 1, 0, 1\}$

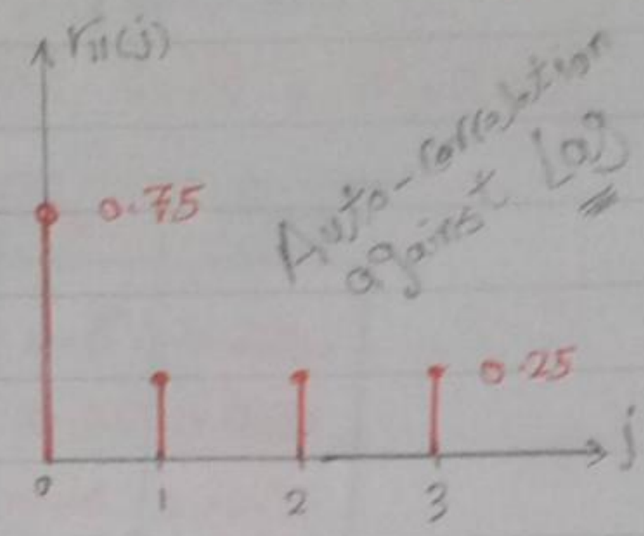
Auto Correlation means signal and itself

$$r_{11}(j) = \frac{1}{N} \sum_{n=0}^{N-1} X_1(n)\, X_1(n+j) \quad , \quad j = 0, \ldots N-1$$

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j	$X_1(n)$	1	1	0	1	$r_{11}(j)$
0	$X_1(n+0)$	(1)	1	0	1	0.75
1	$X_1(n+1)$	1	0	1	0	0.25
2	$X_1(n+2)$	0	1	0	0	0.25
3	$X_1(n+3)$	1	0	0	0	0.25

Auto-Correlation
"Shift Left only"



at $j=0$

$$r_{11}(0) = \frac{1}{N}\sum_{n=0}^{N-1} X_1(n)\,X_1(n+j)$$

$$= \frac{1}{4}\left[X_1(0)X_1(0) + X_1(1)X_1(1) + X_1(2)X_1(2) + X_1(3)X_1(3)\right]$$

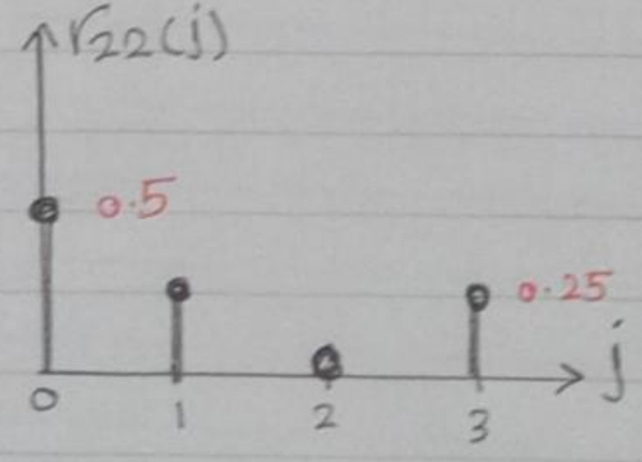
$$= \frac{1}{4}\left[1\cdot1 + 1\cdot1 + 0\cdot0 + 1\cdot1\right] = \frac{3}{4} = 0.75$$

---

* Auto-Correlation For $X_2(n) = \{1, 0, 0, 1\}$

$$r_{22}(j) = \frac{1}{N}\sum_{n=0}^{N-1} X_2(n)\,X_2(n+j) \quad , \quad j = 0 \cdots N-1$$

j	$X_2(n)$	1	0	0	1	$r_{22}(j)$
0	$X_2(n+0)$	1	0	0	1	0.5
1	$X_2(n+1)$	0	0	1	0	0.25
2	$X_2(n+2)$	0	1	0	0	0
3	$X_2(n+3)$	1	0	0	0	0.25



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* Cross-Correlation

$$r_{12}(j) = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n)\, x_2(n+j), \quad j = 0, \ldots, N-1$$

Shift Left + Rotate

j	$x_1(n)$	1	1	0	1	$r_{12}(j)$	$\rho_{12}(j)$
0	$x_2(n+0)$	(1)	0	0	1	0.5	0.8196
1	$x_2(n+1)$	(0)	0	1	(1)	0.25	0.4098
2	$x_2(n+2)$	0	1	1	(0)	0.25	0.4098
3	$x_2(n+3)$	1	1	0	0	0.5	0.8196

* The cross-correlation coefficient

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\sqrt{\sum_{n=0}^{N-1} x_1(n)^2 \cdot \sum_{n=0}^{N-1} x_2(n)^2}}$$

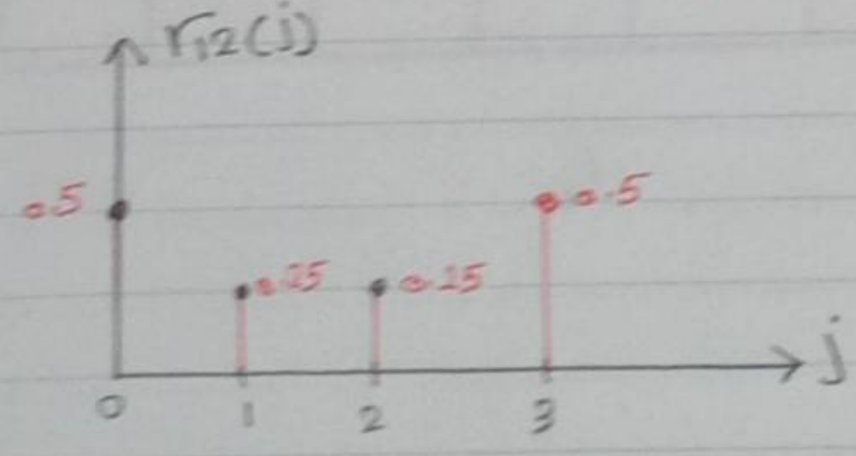
$$\sum_{n=0}^{N-1} x_1(n)^2 = 1^2+1^2+0^2+1^2 = 3 \qquad \sum x_2(n)^2 = 1^2+0^2+0^2+1^2 = 2$$

$$\sum x_1(n)^2 \cdot \sum x_2(n)^2 = 6 \qquad \sqrt{\sum x_1(n)^2 \cdot \sum x_2(n)^2} = \sqrt{6}$$

$$\frac{1}{N}\sqrt{\sum x_1(n)^2 \cdot x_2(n)^2} = \frac{1}{4} \cdot \sqrt{6} = \frac{\sqrt{6}}{4} = 0.61$$

$$\rho_{12}(0) = \frac{r_{12}(0)}{0.61} = \frac{0.5}{0.61} = 0.819$$

* Plot the cross-correlation against lag



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(3.5) Given The Two sets of data

$x_1(n) = \{1.5, 2, 1.5, 2, 2.5\}$ & $x_2(n) = \{0, 0.33, 0.67, 1\}$

Calculate The Following

① Auto Correlation of the Two sequance

② cross-Correlation using direct method.

③ Find cross-Correlation Coefficient.

④ Plot the cross-Correlation against Lag.

solution

* Auto-Correlation For $x_1(n)$ $N_1 = 5$ samples

$$r_{11}(j) = \frac{1}{N} \sum_{n=0}^{N-} x_1(n)\, x_1(n+j)$$

j	$x_1(n)$	1.5	2	1.5	2	2.5	$r_{11}(j)$
0	$x_1(n+0)$	1.5	2	1.5	2	2.5	3.75
1	$x_1(n+1)$	2	1.5	2	2.5	0	2.8
2	$x_1(n+2)$	1.5	2	2.5	0	0	2
3	$x_1(n+3)$	2	2.5	0	0	0	1.6
4	$x_1(n+4)$	2.5	0	0	0	0	0.75

* For Auto Correlation $x_2(n)$ $N_2 = 4$ samples

j	$x_2(n)$	0	0.33	0.67	1	$r_{22}(j)$
0	$x_2(n+0)$	0	0.33	0.67	1	0.389
1	$x_2(n+1)$	0.33	0.67	1	0	0.222
2	$x_2(n+2)$	0.67	1	0	0	0.0825
3	$x_2(n+3)$	1	0	0	0	0

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(2) Cross- Correlation

$N_1 = 5$ samples

$N_2 = 4$ samples

$N_{new} = N_1 + N_2 - 1 = 5 + 4 - 1 = 8$ Called Linear-cross-Cor

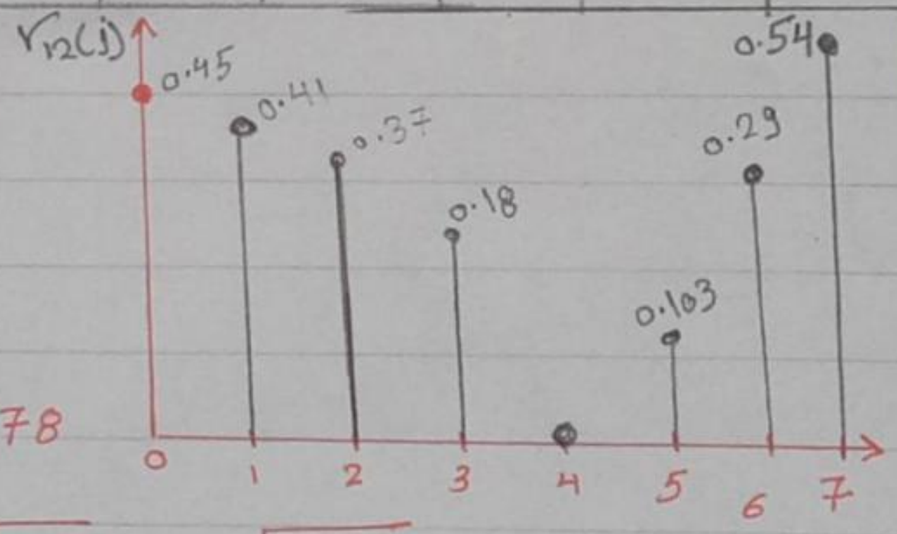
$x_1(n) = \{1.5, 2, 1.5, 2, 2.5, 0, 0, 0\}$

$x_2(n) = \{0, 0.33, 0.67, 1, 0, 0, 0, 0\}$

j	$x_1(n)$	1.5	2	1.5	2	2.5	0	0	0	$r_{12}(j)$	$\rho_{12}(j)$
0	$x_2(n+0)$	0	0.33	0.67	1	0	0	0	0	0.458	0.678
1	$x_2(n+1)$	0.33	0.67	1	0	0	0	0	0	0.4168	0.617
2	$x_2(n+2)$	0.67	1	0	0	0	0	0	0.33	0.3756	0.556
3	$x_2(n+3)$	1	0	0	0	0	0	0.33	0.67	0.1875	0.276
4	$x_2(n+4)$	0	0	0	0	0	0.33	0.67	1	0	0
5	$x_2(n+5)$	0	0	0	0	0.33	0.67	1	0	0.103	0.152
6	$x_2(n+6)$	0	0	0	0.33	0.67	1	0	0	0.2918	0.432
7	$x_2(n+7)$	0	0	0.33	0.67	1	0	0	0	0.5418	0.802

* Cross-Correlation Coefficient

$$\rho_{12}(j) = \frac{r_{12}(j)}{\frac{1}{N}\sqrt{\sum_{n=0}^{N-1} x_1(n)^2 \cdot \sum_{n=0}^{N-1} x_2(n)^2}}$$



$\sum x_1(n)^2 = 18.75$, $\sum_{n=0}^{N-1} x_2(n)^2 = 1.5578$

$\sum x_1(n)^2 \cdot \sum x_2(n)^2 = 29.208$, $\sqrt{\sum x_1(n)^2 \sum x_2(n)^2} = \sqrt{29.08} = 5.4045$

$\rightarrow \frac{1}{N} \cdot 5.4045 = \frac{5.4045}{8} = 0.6755$

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